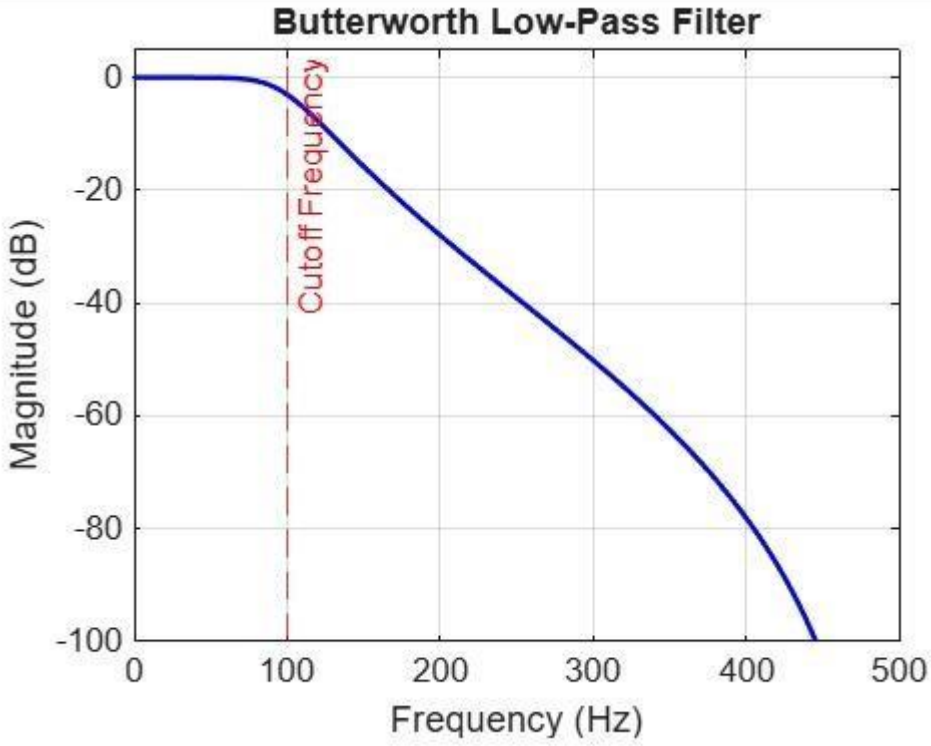
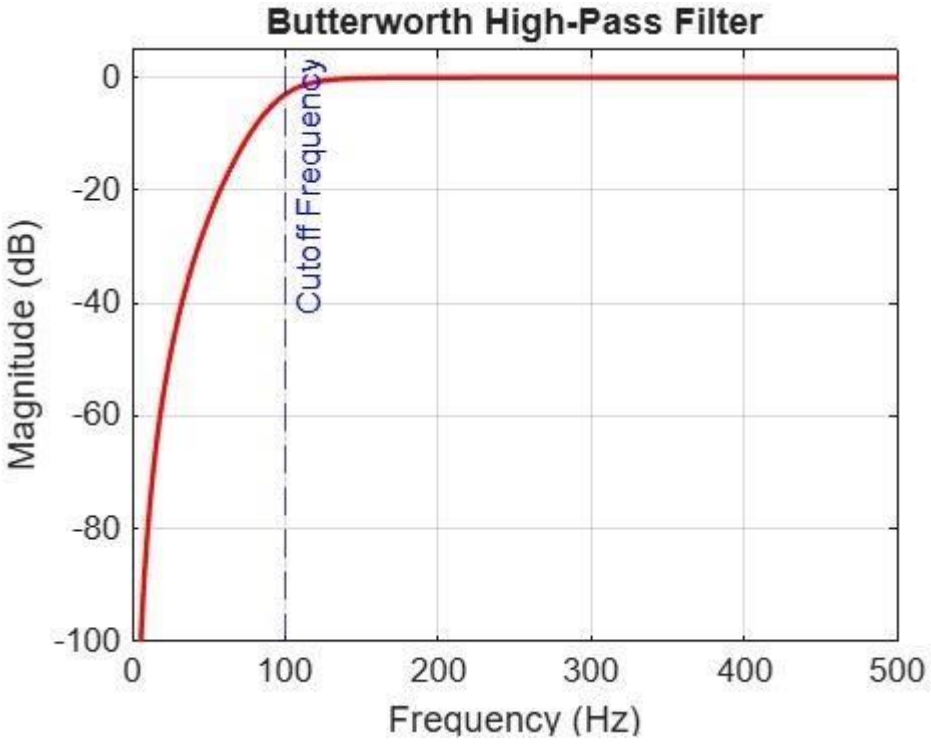
SreeRahul SJ (192312569)

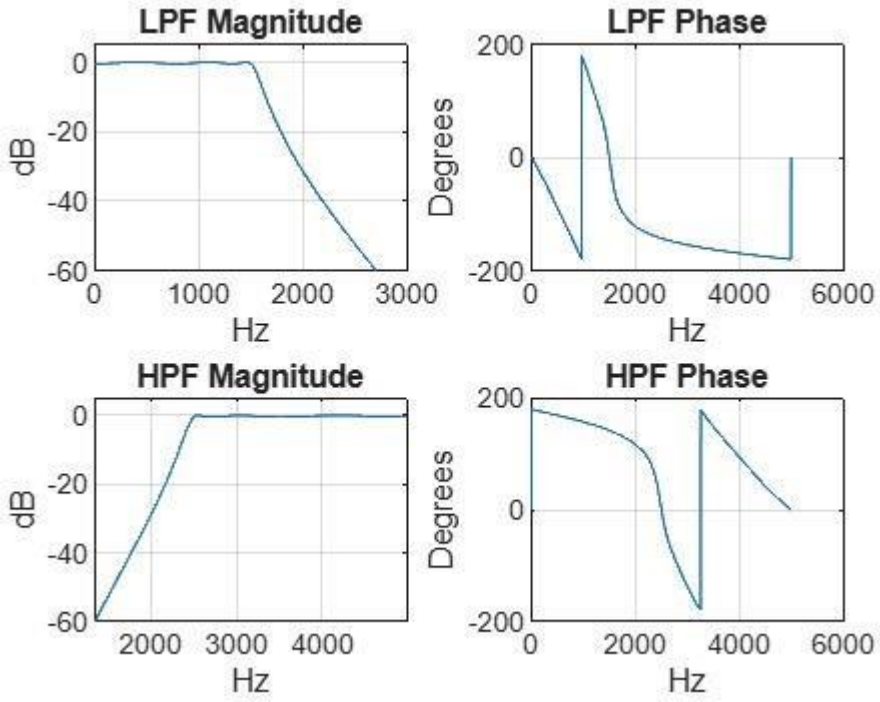
1 TO8 EXPERIMENTS GRAPHS

EXP:1

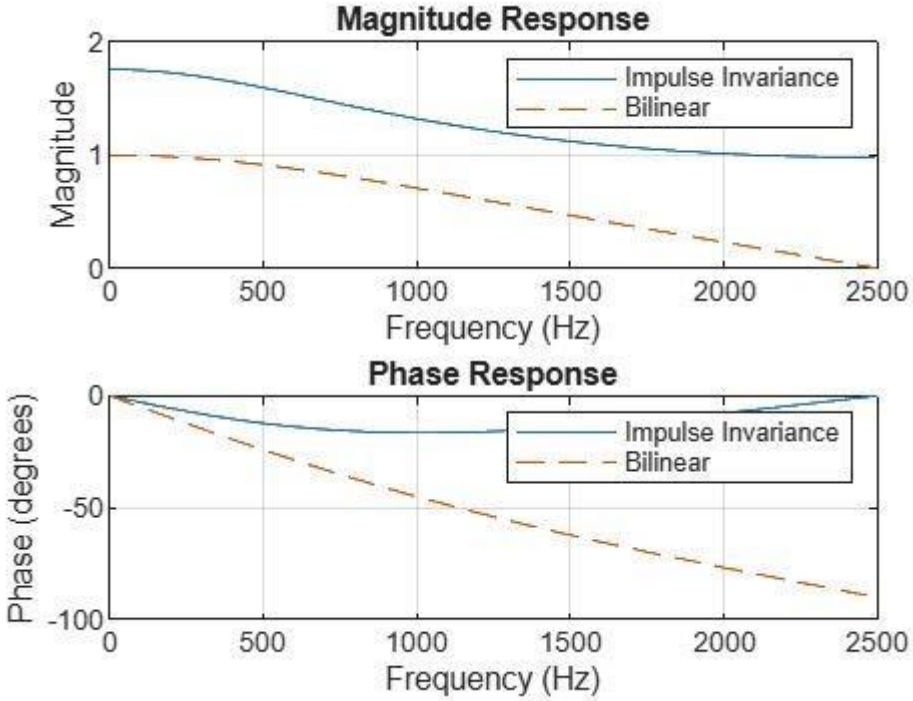




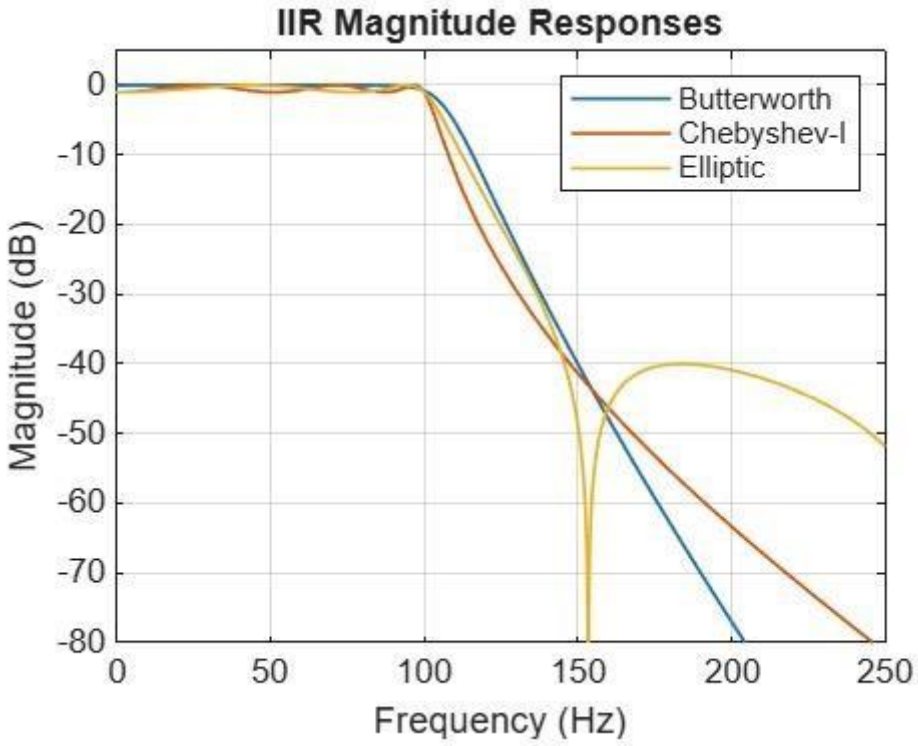
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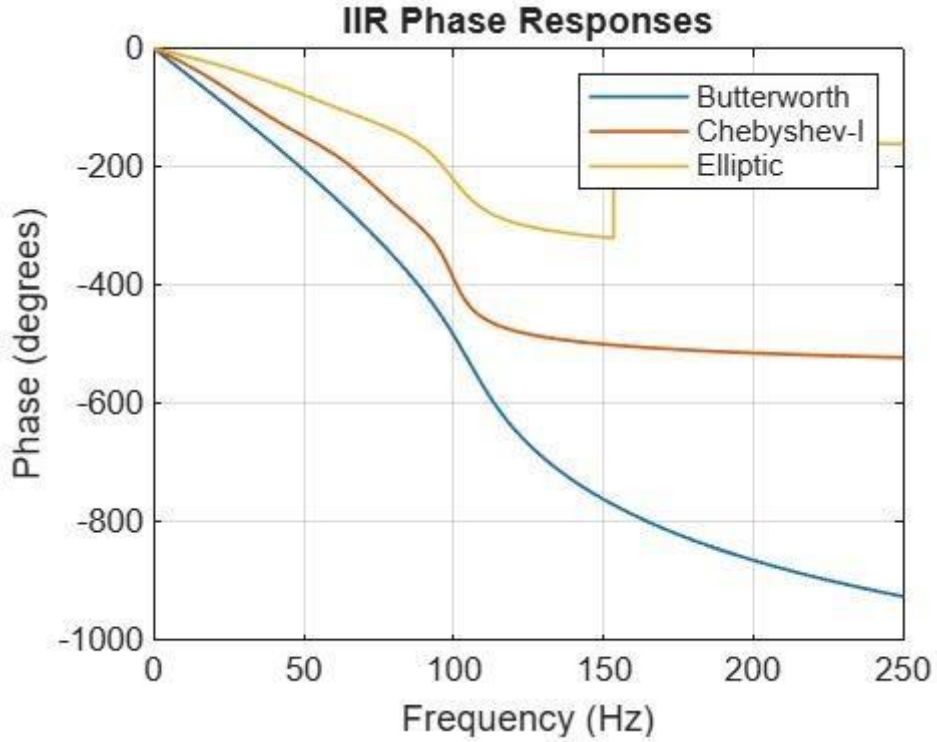

LPF Magnitude
dB
0
-20
-40
-60
0
1000
2000
3000
Hz
LPF Phase
Degrees
200
0
-200
0
2000
4000
6000
Hz
HPF Magnitude
dB
0
-20
-40
-60
2000
3000
4000
Hz
HPF Phase
Degrees
200
0
-200
0
2000
4000
6000
Hz

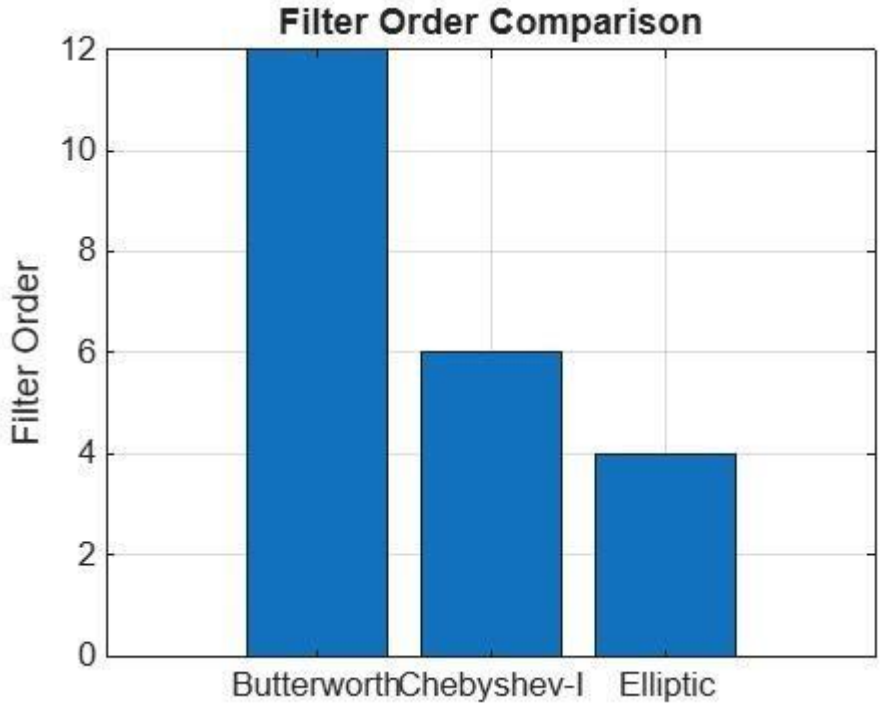
EXP:3


Magnitude Response
Magnitude
2
1
0
0
500
1000
1500
2000
2500
Frequency (Hz)
Impulse Invariance
Bilinear
Phase Response
Phase (degrees)
0
-50
-100
0
500
1000
1500
2000
2500
Frequency (Hz)
Impulse Invariance
Bilinear

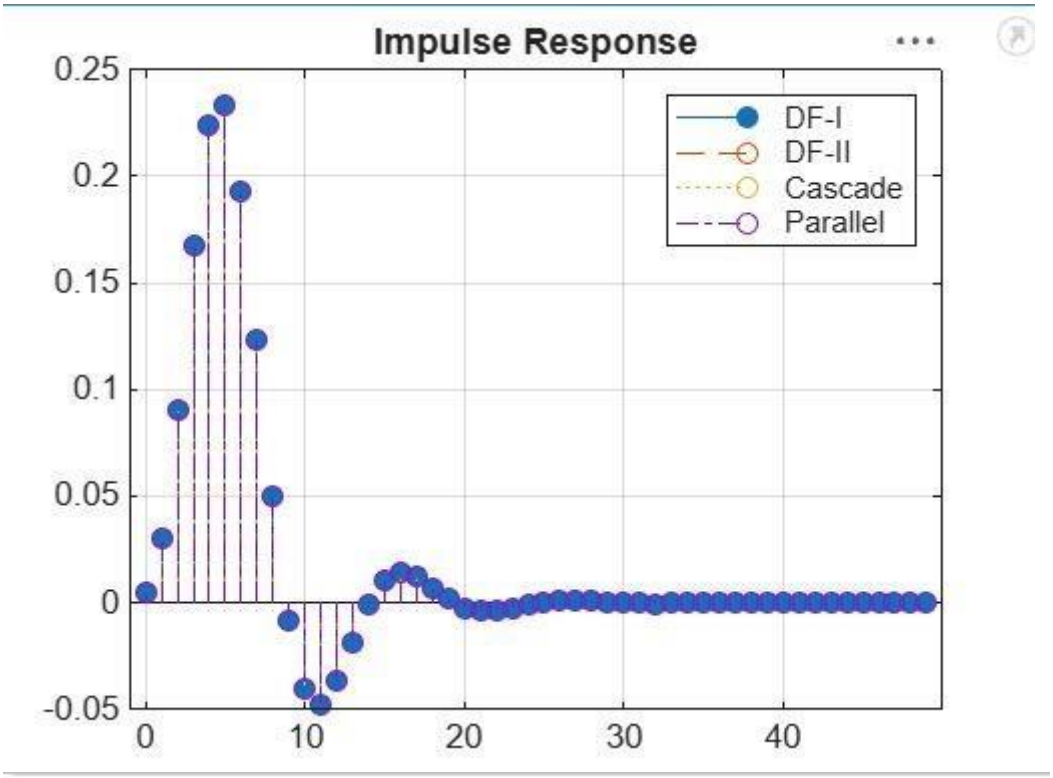
EXP:4

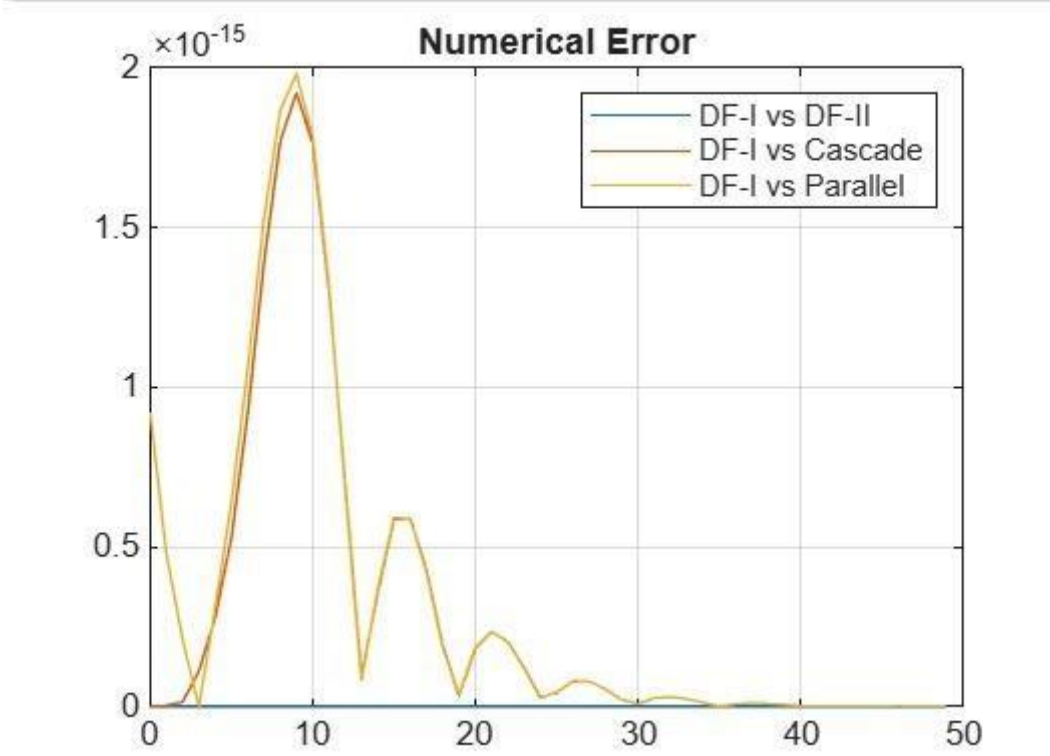

IIR Magnitude Responses
Butterworth
Chebyshev-I
Elliptic
Magnitude (dB)
0
-10
-20
-30
-40
-50
-60
-70
-80
0
50
100
150
200
250
Frequency (Hz)

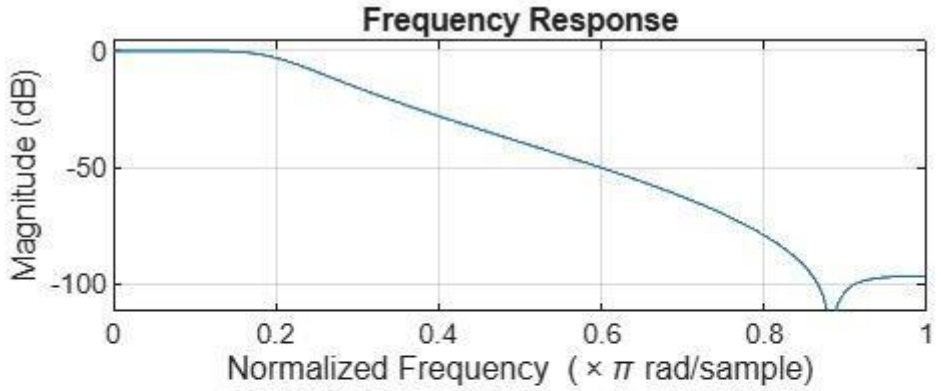

IIR Phase Responses
Butterworth
Chebyshev-I
Elliptic
Phase (degrees)
0
-200
-400
-600
-800
-1000
0
50
100
150
200
250
Frequency (Hz)

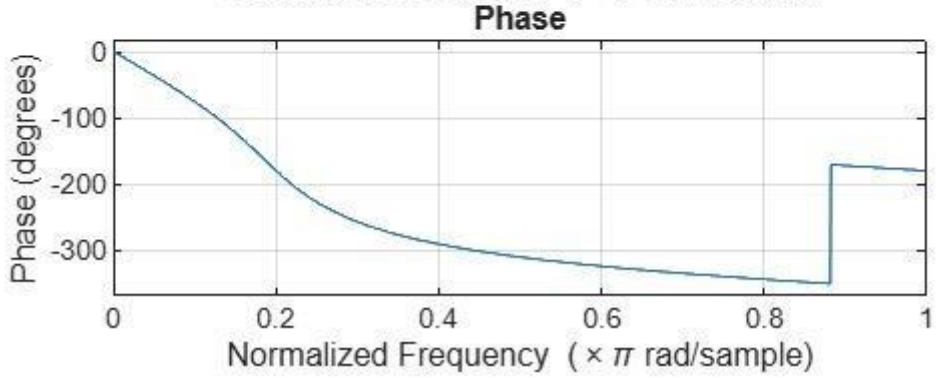

Filter Order Comparison
Filter Order
12
10
8
6
4
2
0
Butterworth
Chebyshev-I
Elliptic

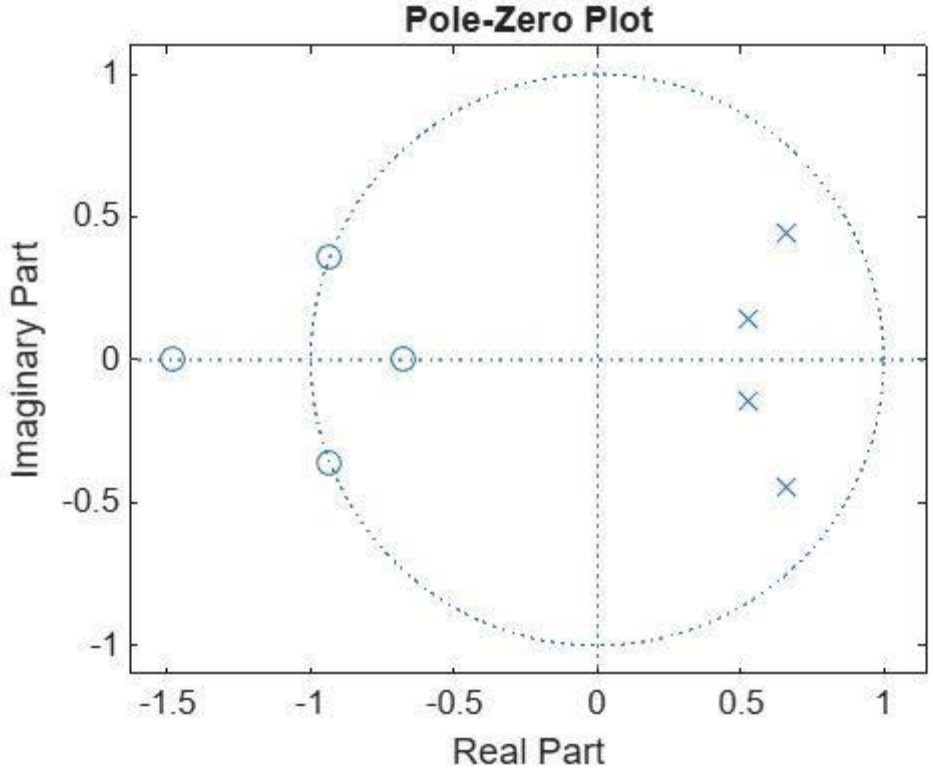
EXP:5


Impulse Response
DF-I
DF-II
Cascade
Parallel
0.25
0.2
0.15
0.1
0.05
0
-0.05
0
10
20
30
40

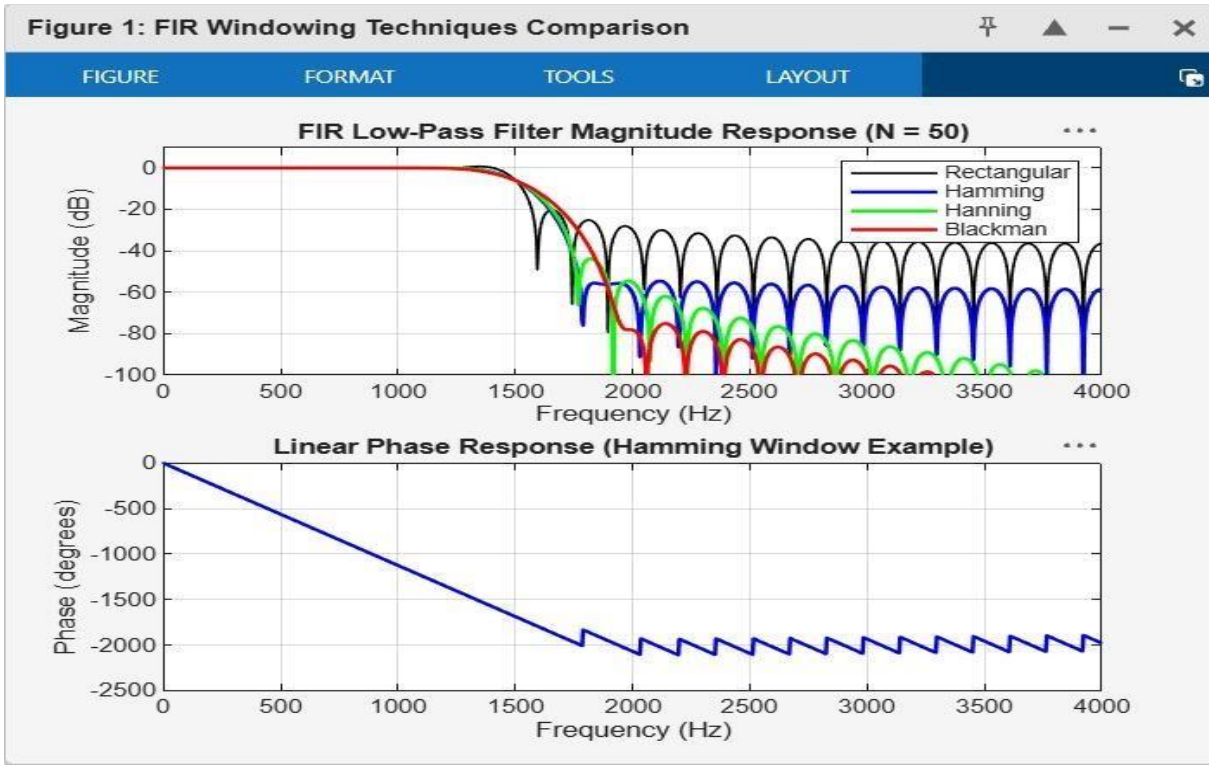

Numerical Error
×10^-15
DF-I vs DF-II
DF-I vs Cascade
DF-I vs Parallel
2
1.5
1
0.5
0
0
10
20
30
40
50


Frequency Response
Magnitude (dB)
0
-50
-100
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)

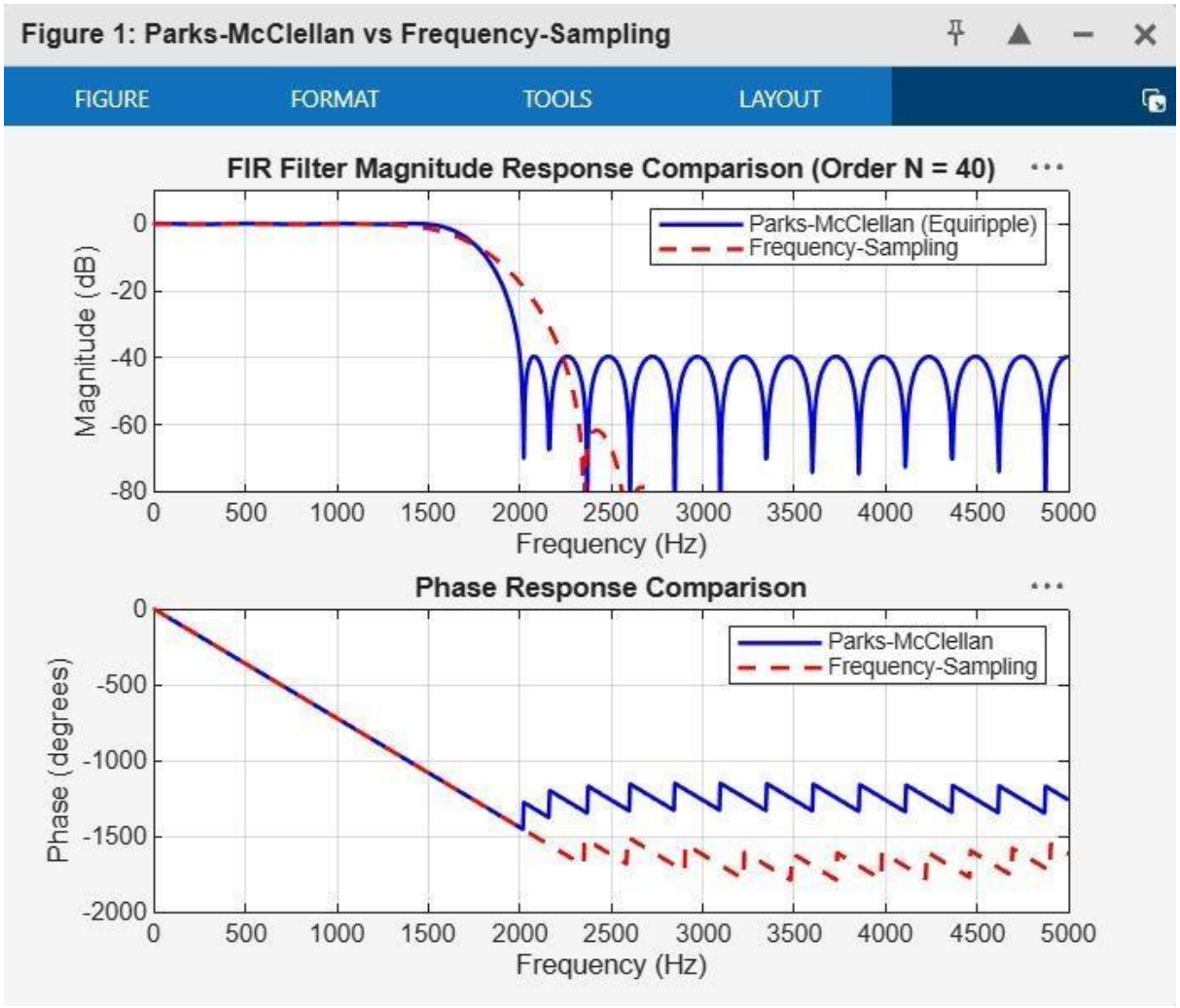

Phase
Phase (degrees)
0
-100
-200
-300
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)


Pole-Zero Plot
Imaginary Part
1
0.5
0
-0.5
-1
-1.5
-1
-0.5
0
0.5
1
Real Part

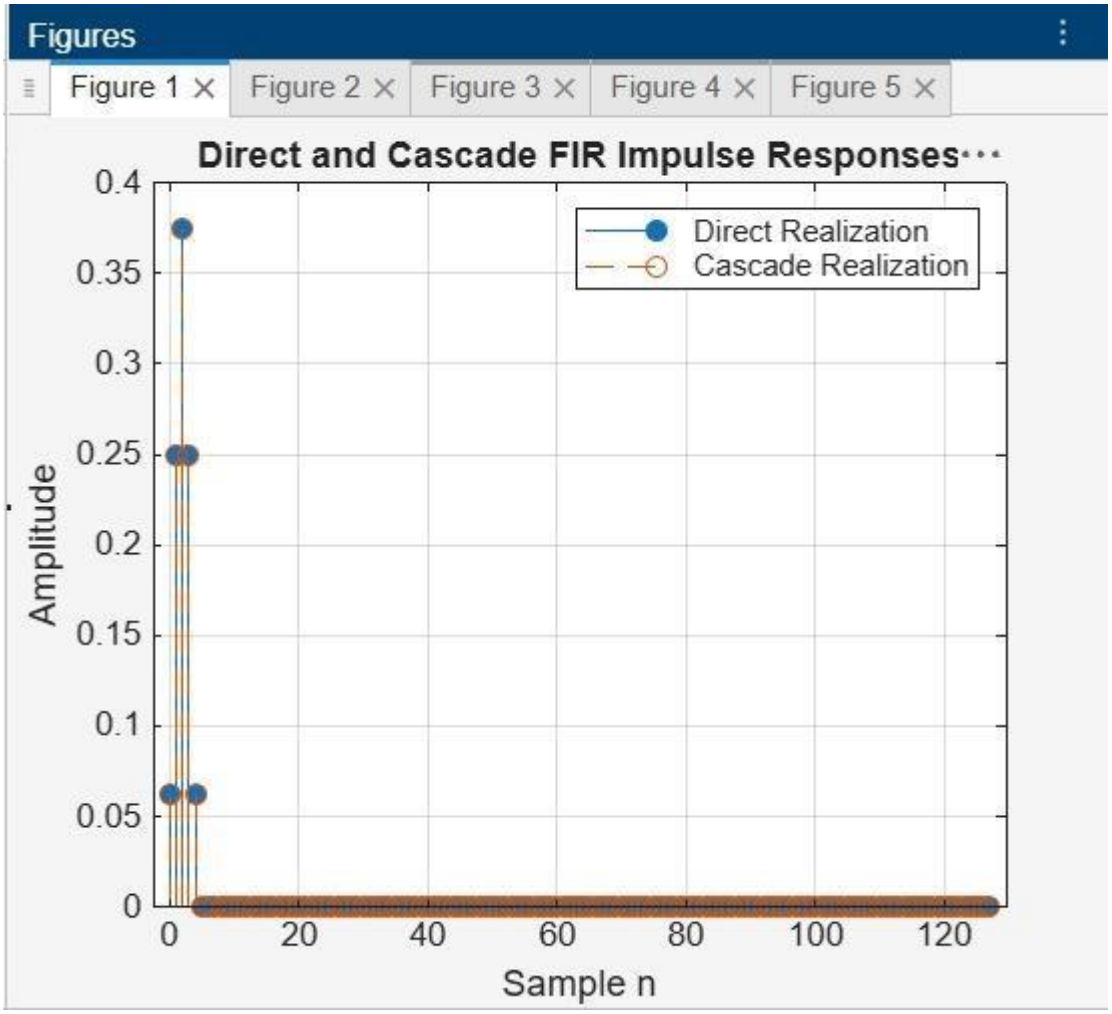
EXP:6

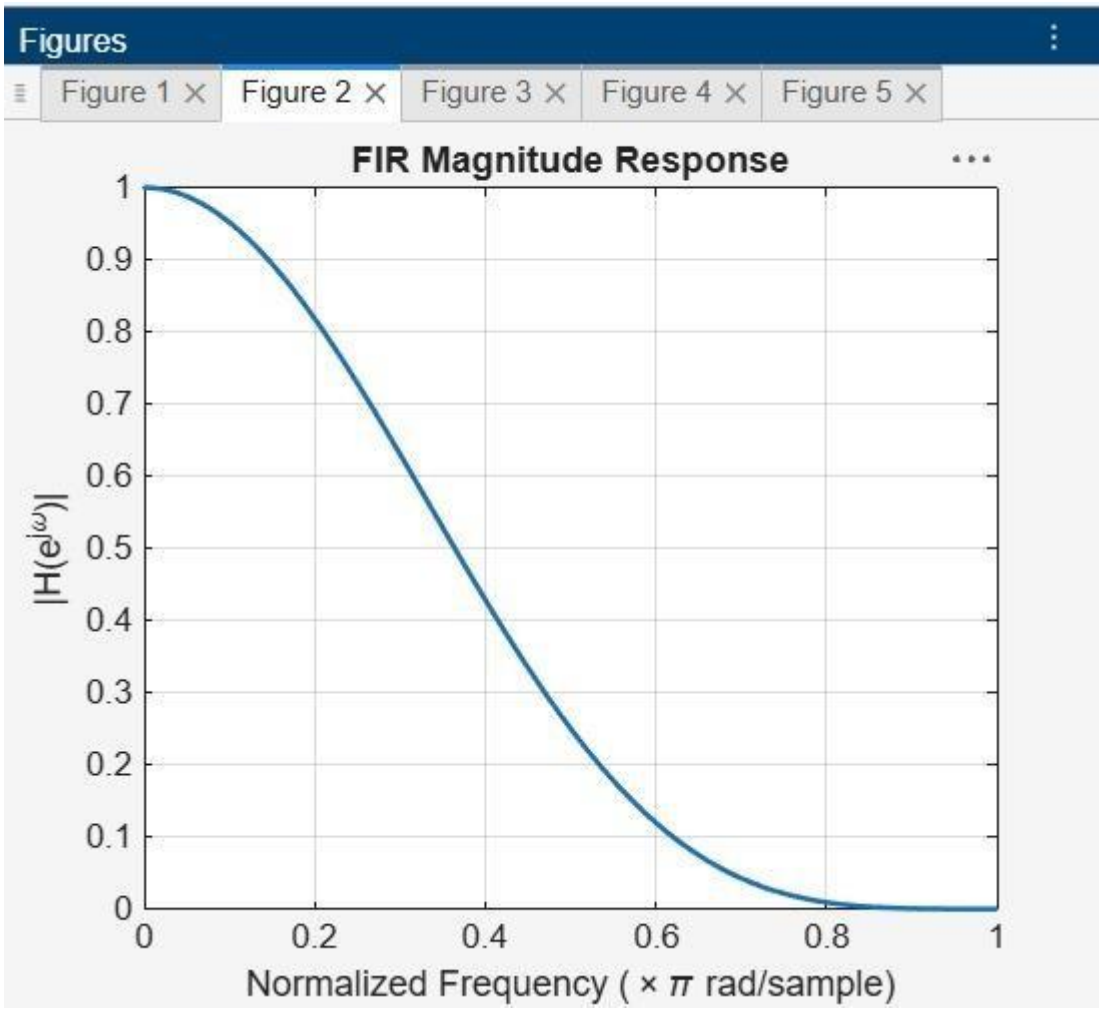

Figure 1: FIR Windowing Techniques Comparison
FIGURE
FORMAT
TOOLS
LAYOUT
FIR Low-Pass Filter Magnitude Response (N = 50)
Magnitude (dB)
Frequency (Hz)
Rectangular
Hamming
Hanning
Blackman
Linear Phase Response (Hamming Window Example)
Phase (degrees)
Frequency (Hz)

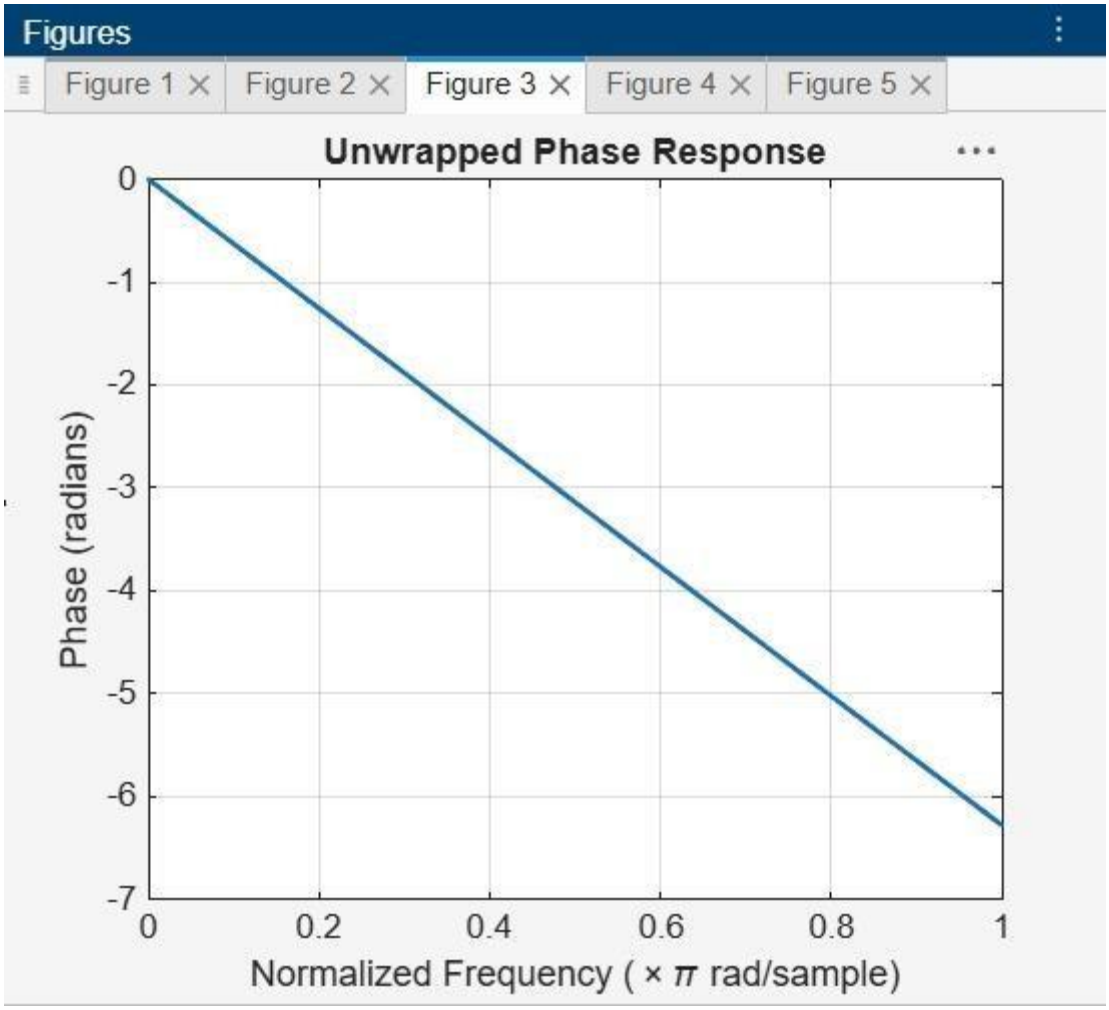
EXP:7

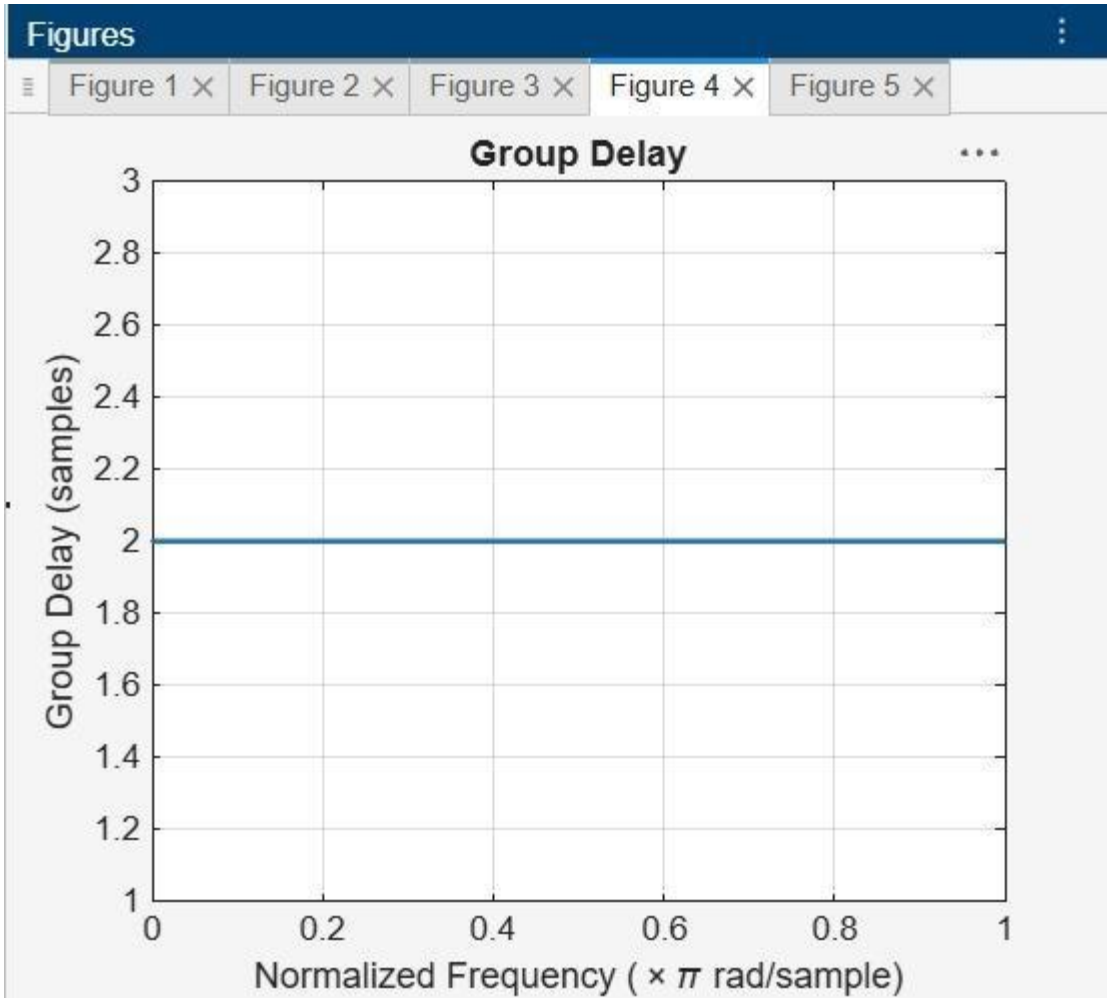

Figure 1: Parks-McClellan vs Frequency-Sampling
FIGURE
FORMAT
TOOLS
LAYOUT
FIR Filter Magnitude Response Comparison (Order N = 40)
Parks-McClellan (Equiripple)
Frequency-Sampling
Magnitude (dB)
0
-20
-40
-60
-80
0
500
1000
1500
2000
2500
3000
3500
4000
4500
5000
Frequency (Hz)
Phase Response Comparison
Parks-McClellan
Frequency-Sampling
Phase (degrees)
0
-500
-1000
-1500
-2000
0
500
1000
1500
2000
2500
3000
3500
4000
4500
5000
Frequency (Hz)

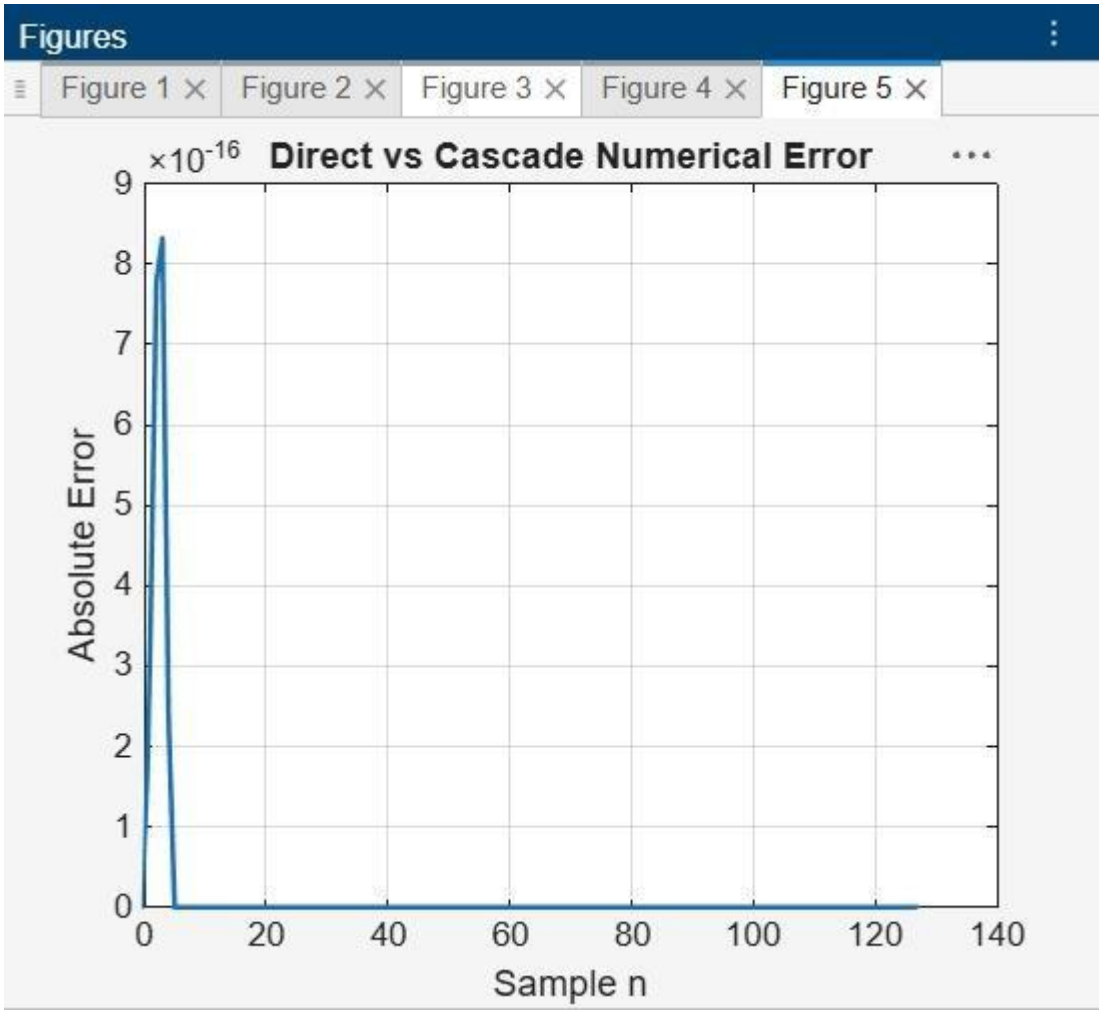
EXP:8


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Direct and Cascade FIR Impulse Responses
Direct Realization
Cascade Realization
Amplitude
Sample n
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0
20
40
60
80
100
120


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
FIR Magnitude Response
|H(e^jω)|
Normalized Frequency (× π rad/sample)
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
0.2
0.4
0.6
0.8
1


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Unwrapped Phase Response
Phase (radians)
0
-1
-2
-3
-4
-5
-6
-7
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Group Delay
Group Delay (samples)
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
1.2
1
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)


Figures
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Direct vs Cascade Numerical Error
×10^-16
Absolute Error
Sample n
0
1
2
3
4
5
6
7
8
9
0
20
40
60
80
100
120
140